

Differential Forms and Currents

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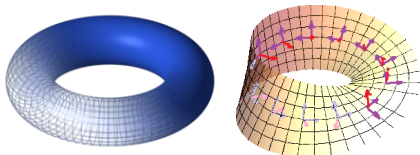
2 Differential Forms

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Motivation

Why Differential Forms?

- Measures let us integrate over spaces, but without a sense of "orientation": Integrating a function "forward" over the interval $[a, b]$ is the same as "backward."
- Differential forms let us integrate over oriented manifolds: surfaces where we can define a consistent notion of "clockwise".
 - ex. planes, spheres, tori
- Applications: Generalized Stokes' Theorem



Background

Definition

The set of m -vectors $\wedge_m \mathbb{R}^n$ is the vector space of formal linear combinations of symbols of the form $v_1 \wedge v_2 \wedge \dots \wedge v_m$, for $v_i \in \mathbb{R}^n$, under the following relations:

$$\begin{aligned} c(v_1 \wedge \dots \wedge v_m) &= v_1 \wedge \dots \wedge cv_i \wedge \dots \wedge v_m \\ v_1 \wedge \dots v_i \dots v_j \dots \wedge v_m &= -(v_1 \wedge \dots v_j \dots v_i \dots \wedge v_m) \end{aligned}$$

for all $c \in \mathbb{R}$, $1 \leq i < j \leq m$, $v_k \in \mathbb{R}^n$, and

$$(v_1 \wedge v_2 \wedge \dots \wedge v_m) + (u_1 \wedge v_2 \wedge \dots \wedge v_m) = (v_1 + u_1) \wedge v_2 \wedge \dots \wedge v_m.$$

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Remark

The second relation above implies that if $v_i = v_j$ for $i \neq j$, the element $v_1 \wedge v_2 \wedge \dots \wedge v_m = 0$.

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Example

$\wedge_1 \mathbb{R}^n \simeq \mathbb{R}^n$ for all n .

Background

Remark

Given the standard basis e_i , $1 \leq i \leq n$ of $\mathbb{R}^n$, a basis of $\wedge_m \mathbb{R}^n$ is the set of m -vectors of the form

$$e_{i_1} \wedge e_{i_2} \wedge \dots \wedge e_{i_m}$$

for all $1 \leq i_1 < i_2 < \dots < i_m \leq n$.

Background

Definition

The set of m -covectors $\wedge^m \mathbb{R}^n$ is the vector space of functions from $\wedge_m \mathbb{R}^n$ to $\mathbb{R}$ (the dual space of $\wedge_m \mathbb{R}^n$). This has a basis of elements

$$dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_m},$$

where the above is defined on a basis element of $e_{j_1} \wedge e_{j_2} \wedge \dots \wedge e_{j_m}$ of $\wedge_m \mathbb{R}^n$ to be 1 if the sequences i_k and j_k are equal and 0 otherwise.

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Example

$\wedge^1 \mathbb{R}^n$ is the space of linear functions from $\mathbb{R}^n$ to $\mathbb{R}$. This has a basis of the coordinate functions dx_i , where $dx_i((a_1, a_2, \dots, a_n)) = a_i$.

Background

Remark

$\wedge^m \mathbb{R}^n$ can also be viewed as the set of alternating functions on $(\mathbb{R}^n)^m$, i.e. functions $\phi : (\mathbb{R}^n)^m \rightarrow \mathbb{R}$ such that $\phi((v_1, \dots, v_m)) = 0$ whenever $v_i = v_j$ for $i \neq j$.

With this motivation, we can also construct elements of $\wedge^m \mathbb{R}^n$ in the following manner:

Background

Definition

Let $f_1, \dots, f_m \in \wedge^1 \mathbb{R}^n$. Using the basis defined previously, we write $f_i = f_{i,1}dx_1 + \dots + f_{i,n}dx_n$. We define the **exterior product** $f_1 \wedge \dots \wedge f_m \in \wedge^m \mathbb{R}^n$ by

$$(f_1 \wedge \dots \wedge f_m)(v_1, \dots, v_m) \\ = \det \begin{pmatrix} f_{1,1} & f_{1,2} & \dots & f_{1,n} \\ f_{2,1} & f_{2,2} & \dots & f_{2,n} \\ \vdots & & & \vdots \\ f_{m,1} & f_{m,2} & \dots & f_{m,n} \end{pmatrix} \begin{pmatrix} v_{1,1} & v_{1,2} & \dots & v_{1,m} \\ v_{2,1} & v_{2,2} & \dots & v_{2,m} \\ \vdots & & & \vdots \\ v_{n,1} & n_{n,2} & \dots & n_{n,m} \end{pmatrix}$$

We can check that this is an m -linear, alternating function.

Differential Forms

We're now ready to define a differential form!

Definition

Let $U \subseteq \mathbb{R}^n$ be an open set. A differential m -form on U , or a differential form with degree m , is a function $\phi : U \rightarrow \wedge^m \mathbb{R}^n$.

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Example

Define $DX_i : \mathbb{R}^n \rightarrow \wedge^1 \mathbb{R}^n$ by $DX_i(x) = dx_i$ for all $x \in \mathbb{R}^n$. This is a differential form of degree 1 on $\mathbb{R}^n$.

Differential Forms

Another way to view a differential form:

Remark

Let $\phi : U \rightarrow \wedge^m \mathbb{R}^n$ be a differential form. For all m -tuples $\mathbf{i} = (i_1, \dots, i_m)$ where $1 \leq i_1 < i_2 < \dots < i_m \leq m$, we define a function $\phi_{\mathbf{i}} : U \rightarrow \mathbb{R}$ by

$$\phi_{\mathbf{i}}(u) = \phi(u)(e_{i_1} \wedge \dots \wedge e_{i_m}).$$

Now we can write

$$\phi = \sum_{\mathbf{i}} \phi_{\mathbf{i}}(dx_{i_1} \wedge \dots \wedge dx_{i_m}),$$

since the basis $\{dx_{i_1} \wedge \dots \wedge dx_{i_m}\}$ is the dual basis to $\{e_{i_1} \wedge \dots \wedge e_{i_m}\}$.

Integral of a Differential Form

Definition

Let $U \subseteq \mathbb{R}^n$ be $\mathcal{L}^n$ -measurable, and let ϕ be a continuous differential n -form on U . We define the integral of ϕ over U :

$$\int_U \phi = \int_U \phi(x)(e_1 \wedge \dots \wedge e_n) d\mathcal{L}^n x.$$

On the left side, U is oriented by with the induced orientation from the orientation of $\mathbb{R}^n$ given by $e_1 \wedge \dots \wedge e_n$.

Currents

Definition

An m -dimensional **current** on U is a linear functional on the space of infinitely differentiable differential m -forms on U with compact support. (i.e. a function mapping a differential form to a real number.)

These are useful for integration over rectifiable sets.

Currents

Example

Let $\xi : U \rightarrow \wedge_m \mathbb{R}^n$ be a $\mathcal{L}^n$ -measurable function such that the operator norm $|\xi| \in L^1(U)$. We define an m -dimensional current T by

$$T(\phi) = \int_U \phi(x)(\xi(x)) d\mathcal{L}^n x.$$

References

H. Federer, *Geometric Measure Theory*, Classics in Mathematics, Springer Verlag Berlin Heidelberg, 1996.

S. Krantz and H. Parks, *Geometric Integration Theory*, Birkhäuser, Boston, 2008.